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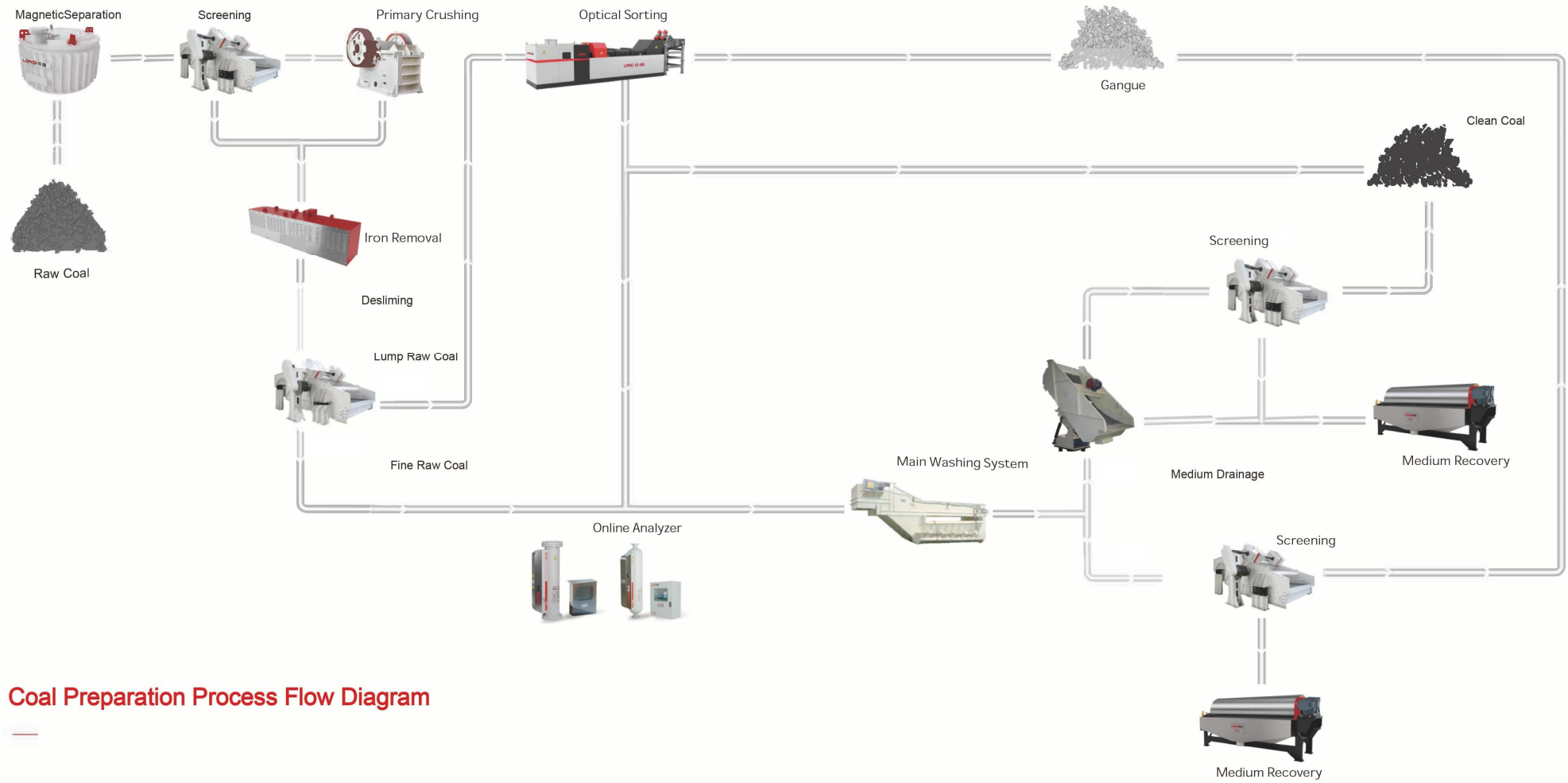
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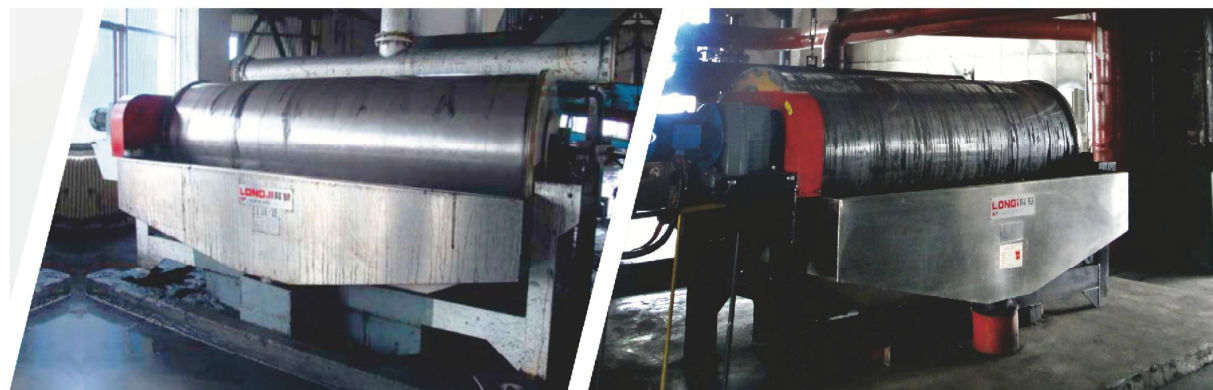
HMDS Heavy Media Recovery Drum Separator

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The most competitive supplier of industrial magnetic
equipment around the world.



Coal Preparation Process Flow Diagram



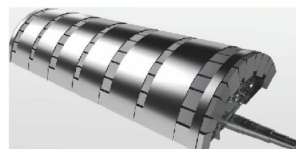
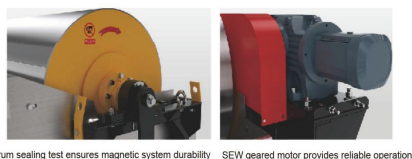
HMDS Series / Wet Counter-Current Magnetic Separator

Application

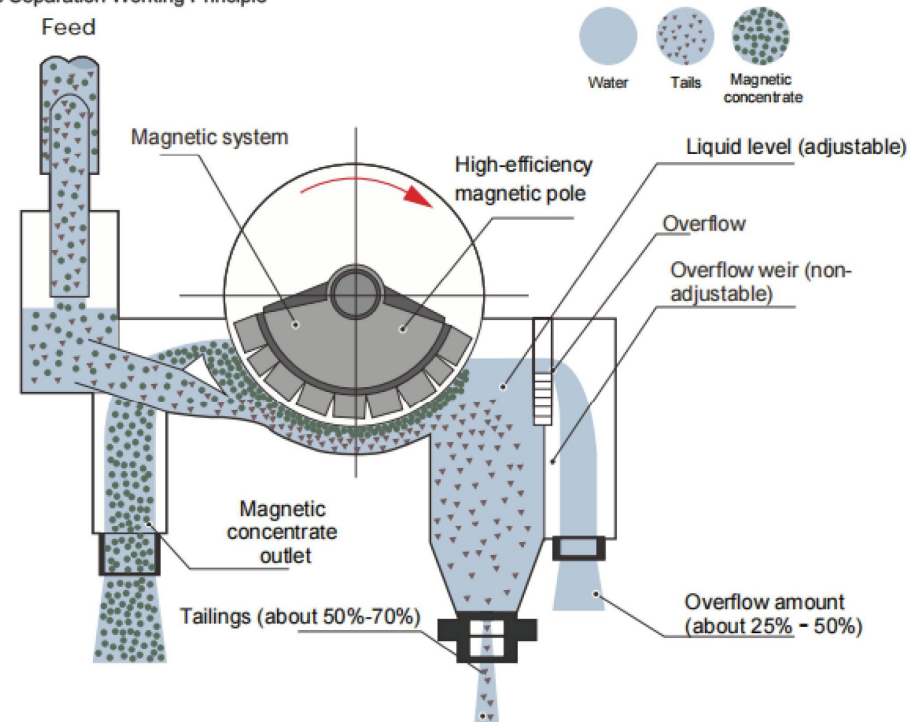
Used for automatic and continuous recovery of ferromagnetic dense media (magnetite or ferrosilicon) in dense-medium separation processes, such as coal, phosphate, fluorite, and other mineral processing plants. Suitable for magnetic material recovery and applications with relatively low magnetic mineral requirements. It is widely used in the coal washing industry and is an important component of wet coal preparation systems. The equipment adopts advanced magnetic circuit design and high-quality magnetic materials, ensuring reliable magnetic separation performance.

Product Advantages

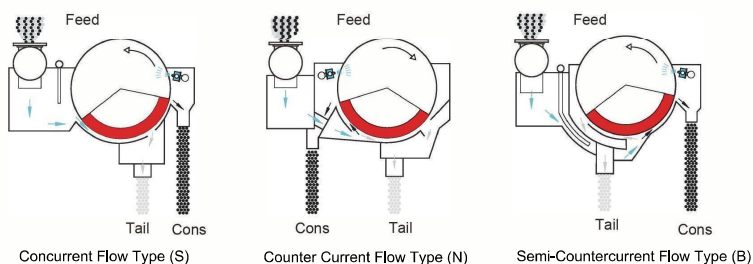
- Magnetic system sealed inside a double-layer wear-resistant stainless steel drum, with additional stainless-steel protection to prevent loosening or detachment.
- Large magnetic coverage angle ensures tailings slurry remains within the magnetic field, preventing medium loss.
- Double-row spherical roller bearings allow inspection and replacement without removing the drum, offering higher load capacity and longer service life.
- SEW reducer and SKF bearings ensure long-term stable operation.
- Integral magnetization technology provides uniform magnetic field.
- Modular structure for easy installation and magnetic angle adjustment.
- Motor protection device ensures the motor operates under rated conditions.



■ Magnetic Separation Working Principle



Three Types of Permanent Magnetic Drum Separators



Main Technical Parameters

Model / Specification	Drum Diameter (mm)	Drum Length (mm)	Length (mm)	Width (mm)	Height (mm)	Motor Power (KW)	Weight (KG)
HMDS-914X1829	914	1829	2757	2108	1585	3	2300
HMDS-914X2134	914	2134	3062	2108	1585	4	2610
HMDS-914X2438	914	2438	3366	2108	1585	4	2890
HMDS-914X2743	914	2743	3671	2108	1585	4	3170
HMDS-914X2972	914	2972	3877	2108	1585	4	3850
HMDS-1219X1829	1219	1829	2817	2459	1895	4	2950
HMDS-1219X2134	1219	2134	3122	2459	1895	5.5	3250
HMDS-1219X2438	1219	2438	3426	2459	1895	5.5	3350
HMDS-1219X2743	1219	2743	3722	2459	1895	5.5	3875
HMDS-1219X2972	1219	2972	3960	2459	1895	5.5	4750

(For reference only)

Process Performance

Model / Specification		Magnetic System	Slurry Capacity (m ³ /h)	Concentrate Capacity (t/h)	Feed Solid Content (%)	Feed Slurry Composition Requirements (Solid Percentage)	
Drum Diameter	Drum Width		Countercurrent Type Recommended Value			Magnetic Material Content (%)	Non-Magnetic Material Content (%)
HMDS 914mm (36")	1829 (72")	≥1800GS Magnetic Arc Coverage 138°	180	35	10%-25%	13%	12%
	2134 (84")		210	40			
	2438 (96")		235	45			
	2743 (108")		265	50			
	2972 (117")		298	58			
HMDS 1219mm (48")	1829 (72")	≥1800GS Magnetic Arc Coverage 138°	215	65	10%-25%	13%	12%
	2134 (84")		250	75			
	2438 (96")		285	85			
	2743 (108")		320	95			
	2972 (117")		355	105			

Heavy Medium Recovery Rate

Under normal operating conditions (medium density 1.0 – 1.26 kg/m³, typically ~1.13 kg/m³), the HMDS magnetic separator can achieve the following recovery performance:

Feed medium content 87 – 130 kg/m³: recovery > 99.7%

Feed medium content 130 – 260 kg/m³: recovery > 99.8%

Feed medium content > 260 kg/m³: recovery > 99.9%